DecodeLabs E-commerce Order Data Cleaning Documentation

Project Overview

This project was completed as part of the DecodeLabs Data Analytics Internship. The objective was to review an e-commerce order dataset for common data quality issues and clean the data where necessary to ensure it was consistent and ready for analysis.

Dataset Information

Source: DecodeLabs Data Analytics Internship

Tool Used: Microsoft Excel

Number of Rows: 1,200

Number of Columns: 14

Dataset Columns

- Order ID
- Date
- Customer ID
- Product
- Quantity
- Unit Price
- Shipping Address
- Payment Method
- Order Status
- Tracking Number
- Items In Cart
- Coupon Code
- Referral Source
- Total Price

Data Quality Assessment

Before making any changes, I reviewed the dataset to identify common data quality issues. The checks included missing values, duplicate records, incorrect data types, date formatting and unnecessary whitespace.

The following observations were made:

- Missing values were found in the **Coupon Code** column.
- No duplicate records were found.
- The **Date** column was already stored in the correct format.
- The **Quantity**, **Unit Price** and **Total Price** columns were not in the appropriate numeric format.
- No visible leading or trailing spaces were found in the text fields.

Data Cleaning Process

1. Handling Missing Values

I used Conditional Formatting to identify blank cells in the dataset. The only missing values were found in the **Coupon Code** column.

Based on the context of the dataset, the blank cells were interpreted as orders where no coupon was applied. To improve consistency and make the data easier to analyse, the blank cells were replaced with **"No Coupon".**

2. Checking for Duplicate Records

The dataset was checked for duplicate records using Excel's duplicate checking feature.

No duplicate records were identified, so no rows were removed.

3. Correcting Data Types

The **Quantity**, **Unit Price** and **Total Price** columns were reviewed and found to be stored in the “General” data format.

These columns were converted to the **Number** data type to ensure they could be used correctly for calculations and further analysis.

4. Validating the Date Column

The **Date** column was checked using Excel's Filter feature to confirm that all entries were recognised as valid dates. The dates were grouped by year, confirming that the column was already formatted correctly.

No changes were required.

5. Removing Unnecessary Whitespace

Although no visible leading or trailing spaces were found in the text columns, the **TRIM()** function was applied to all text fields as a precaution. This helped ensure the dataset was free from hidden spaces that could affect sorting, filtering or analysis.

Summary of Cleaning Activities

Data Quality Check	Action Taken	Outcome
Missing values	Replaced blank Coupon Code cells with "No Coupon"	Missing values addressed
Duplicate records	Checked for duplicates	No duplicates found
Data types	Converted Quantity, Unit Price, and Total Price to Number format	Numeric columns standardised
Date validation	Verified using Excel Filters	No issues found
Whitespace	Applied the TRIM() function to text columns	Text fields cleaned and standardised

Final Dataset

Description	Result
Rows before cleaning	1,200
Rows after cleaning	1,200
Columns before cleaning	14
Columns after cleaning	14
Rows removed	0
Columns removed	0

No rows or columns were removed during the cleaning process because the dataset contained no duplicate records or unnecessary fields.

Conclusion

This project involved reviewing and cleaning an e-commerce order dataset using Microsoft Excel. Missing values in the **Coupon Code** column were replaced with **"No Coupon"** based on the assumption that blank entries indicated no coupon was used. The **Quantity**, **Unit Price**, and **Total Price** columns were converted to the correct numeric data type and the **TRIM()** function was applied to text columns to ensure consistency.

Additional checks confirmed that there were no duplicate records and that the **Date** column was already correctly formatted. After completing these steps, the dataset was clean, consistent and ready for further analysis.